

Anatomy of health effects of Mediterranean diet: Greek EPIC prospective cohort study



Objective To investigate the relative importance of the individual components of the Mediterranean diet in generating the inverse association of increased adherence to this diet and overall mortality. **Design** Prospective cohort study. **Setting** Greek segment of the European Prospective Investigation into Cancer and nutrition (EPIC). **Participants** 23 349 men and women, not previously diagnosed with cancer, coronary heart disease, or diabetes, with documented survival status until June 2008 and complete information on nutritional variables and important covariates at enrolment. **Main outcome measure** All cause mortality. **Results** After a mean follow-up of 8.5 years, 652 deaths from any cause had occurred among 12 694 participants with Mediterranean diet scores 0-4 and 423 among 10 655 participants with scores of 5 or more. Controlling for potential confounders, higher adherence to a Mediterranean diet was associated with a statistically significant reduction in total mortality (adjusted mortality ratio per two unit increase in score 0.864, 95% confidence interval 0.802 to 0.932). The contributions of the individual components of the Mediterranean diet to this association were moderate ethanol consumption 23.5%, low consumption of meat and meat products 16.6%, high vegetable consumption 16.2%, high fruit and nut consumption 11.2%, high monounsaturated to saturated lipid ratio 10.6%, and high legume consumption 9.7%. The contributions of high cereal consumption and low dairy consumption were minimal, whereas high fish and seafood consumption was associated with a non-significant increase in mortality ratio. **Conclusion** The dominant components of the Mediterranean diet score as a predictor of lower mortality are moderate consumption of ethanol, low consumption of meat and meat products, and high consumption of vegetables, fruits and nuts, olive oil, and legumes. Minimal contributions were found for cereals and dairy products, possibly because they are heterogeneous categories of foods with differential health effects, and for fish and seafood, the intake of which is low in this population.